

Design and Analysis of Algorithm

Assignment # 04

Question # 01.

Add a boolean flag (or array) in every pass. If any full pass no relaxation occurs, terminate immediately.

$m+1$ passes because:

- Shortest path with exactly k edges are discovered in pass k
- After reaching pass m , no more relaxation occurs.
- The next pass ($m+1$) detects "no update" and stops.

Modified bellmanford loop

repeat

update = false

for each edge (u, v) in E

if $(d[u] + w(u, v) < d[v])$:

$d[v] = d[u] + w(u, v)$

update = true

until update = false

Question # 2

edges .

$r \rightarrow s (5)$ $r \rightarrow t (3)$

$s \rightarrow t (2)$ $s \rightarrow x (6)$

$t \rightarrow x (7)$ $t \rightarrow y (4)$

$x \rightarrow y (-1)$ $x \rightarrow z (1)$

$y \rightarrow z (-2)$

Topological order

r, s, t, x, y, z

At r , relax $r \rightarrow s$ $d[s] = 5$
relax $r \rightarrow t$ $d[t] = 3$

Date: _____

At s: relax $s \rightarrow t(2) = 5+2=7 > 3$ No change.
relax $s \rightarrow x$ $d[x] = 11$

At t: relax $t \rightarrow x(3+7)=10 < 11 \rightarrow d[x] = 10$
relax $t \rightarrow y \rightarrow d[y] = 7$

At x: relax $x \rightarrow y = 10-1=9 > 7$ No change
relax $x \rightarrow z = d[z] = 11$

At y: relax $y \rightarrow z = 7-2=5 < 11 \rightarrow d[z] = 5$

Overall Complexity

Topological sort : $O(V+E)$

Relaxation $O(E)$

Total = $O(V+E)$

Question #03

Dijkstra from s

$d[s] = 0$ others = ∞

$S = \emptyset$

Iterations	Extract	S now	d-values
1	s	{s}	$t=10$ $y=5$ $x=2$
2	x	{s, x}	$z=8$
3	y	{s, x, y}	$t=7$ (via y), $z=7$
4	t	{s, x, y, t}	x improves to 16
5	z	{s, x, y, t, z}	done

Dijkstra from z

$z \rightarrow x(4)$

All others reachable distances

$z = 0$

$x = 4$

$s, t, y = \infty$

Question #04

From	To	Distance	Path
1	6	-?	$1 \rightarrow 4 \rightarrow 2 \rightarrow 6$
2	1	1	$2 \rightarrow 1$
3	1	3	$3 \rightarrow 2 \rightarrow 1$
4	1	-4	$4 \rightarrow 1$
5	1	-1	$5 \rightarrow 1$
6	1	13	$6 \rightarrow 3 \rightarrow 2 \rightarrow 1$

No negative cycle exists

Question #05

DFS starts at B

Vertex	d	f	b) Edge Classification
B	1	18	Using DFS times
C	2	13	
F	3	12	Tree edges
I	4	11	$B \rightarrow C, C \rightarrow F, F \rightarrow I, I \rightarrow H,$
H	5	10	$H \rightarrow G, G \rightarrow D, B \rightarrow E, E \rightarrow A$
G	6	9	
D	7	8	Back edges
E	14	17	$H \rightarrow I$
A	15	16	$E \rightarrow A$ Not a back edge

Forward edge

$A \rightarrow B, D \rightarrow H$

Cross edges

$B \rightarrow E, H \rightarrow F$

Question #06

Max Spanning Tree
Starts at A

1) $A \rightarrow B (7)$ Final Maximum Spanning Tree Weight

2) $A \rightarrow C (5)$ $7 + 5 + 9 + 8 + 4 + 3 = 36$

3) $B \rightarrow D (9)$

4) $D \rightarrow G (8)$

5) $C \rightarrow F (4)$

6) $D \rightarrow E (3)$

b) Yes, if there are multiple edges with equal weight, both algorithms may choose different edges among ties but all resulting trees have the same total weight.

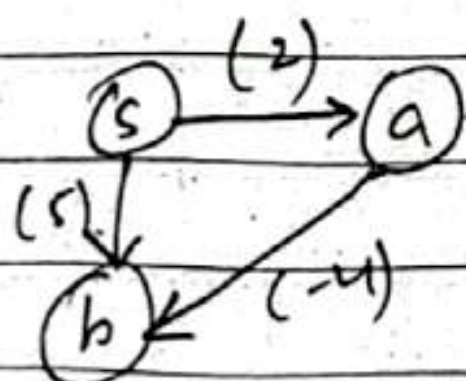
Question #07

(a) Example where Dijkstra fails

$S \rightarrow a (2)$

$S \rightarrow b (5)$

$a \rightarrow b (-4)$



Dijkstra outputs $S \rightarrow b$, which is wrong
 $= 5$

(b) Adding a constant k to all edges weight, it is invalid for example:

Having the same graph

Add $k = 10$

$S \rightarrow a$ becomes 12

$a \rightarrow b$ " 6

$S \rightarrow b$ " 15

(It changes path rankings)

Now path $S \rightarrow a \rightarrow b = 18$

direct $S \rightarrow b = 15$ (incorrectly chosen)

c) If only edges leaving s are negative, Dijkstra still fails

Example $S \rightarrow a (-10)$ $S \rightarrow b (1)$ $a \rightarrow b (15)$

Correct

$S \rightarrow a \rightarrow b = 5$

□ Dijkstra picks b first, never discovers better path via a . Sandal